A WEAKLY ADMISSIBLE RING TOGETHER WITH A WEAK IDEAL OF DEFINITION IS A HENSELIAN PAIR

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The criterion for a henselian pair in terms of lifting maps from etale algebras. More on Algebra, Lemma more-algebra-lemma-characterize-henselian-pair, invoked at proof lines 1124-1126. Reduces henselianity to the lifting of algebra maps.
- Lifting of algebra maps along etale maps modulo a locally nilpotent ideal. More on Algebra, Lemma more-algebra-lemma-locally-nilpotent-henselian, invoked at proof lines 1129-1131. Gives the lifting modulo each open ideal contained in the chosen one.
- Weakly admissible rings are complete, and weak ideals of definition have locally nilpotent quotients by open subideals. Same chapter, the definitions in force in the statement. Supplies completeness and the nilpotence hypothesis.

Source reference. Original lemma 09XI: sources/stacks-chow/tags.txt, source lines 12531–12531. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0ALI: sources/stacks-chow/tags.txt, source lines 13371–13371. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a weakly admissible topological ring. Let $I \subset A$ be a weak ideal of definition. Then (A, I) is a henselian pair.*

Proof. Let $A \to A'$ be an étale ring map and let $\sigma : A' \to A/I$ be an A-algebra map. By More on Algebra, Lemma 09XI it suffices to lift σ to an A-algebra map $A' \to A$. To do this, as A is complete, it suffices to find, for every open ideal $J \subset I$, a unique

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A-algebra map $A' \to A/J$ lifting σ. Since I is a weak ideal of definition, the ideal I/J is locally nilpotent. We conclude by More on Algebra, Lemma 0ALI. $\square$